

PORTFOLIO

Jungin Kim

jungin3486@gmail.com

CONTENTS

1. Experiences
2. Publications
3. Patents
4. Projects

Experiences

EDUCATION

Sep 2023 - Aug 2025 Yonsei University

M.S. in Computer Science

Theory of Computation Lab

GPA: 4.01/4.30

Advisor: Professor Yo-Sub Han

Research Area: Code Watermarking

Mar 2017 - Feb 2022 Sangmyung University

B.S. in Computer Software

Network Data Science Lab

GPA: 3.92/4.50

Advisor: Professor Hyunchul Kim

Research Area: Hate Speech Detection

CAREER

Aug 2023 – Present Yonsei University

Research Assistant, Theory of Computation Lab

Advisor: Professor Yo-Sub Han

Mar 2022 – Apr 2023 Alookso, Inc

Data Analyst, Data Team

Aug 2020 – Nov 2020 KAIST

Research Assistant, Statistical Artificial Intelligence Lab

Advisor: Professor Jaesik Choi

Mar 2019 – Jul 2020 Sangmyung University

Research Intern, Network Data Science Lab

Advisor: Professor Hyunchul Kim

Aug 2017 – Aug 2017 Joycity, Inc

Data Analyst Intern, Kir Team (3on3 FreeStyle)

Experiences

AWARDS & HONORS

Apr 2023 ADsP (Associate Data Analyst Certificate), Korea Data Agency

Dec 2021 - Feb 2022 Completed AIFEL AI Bootcamp, Modu Research Institute

Aug 2021 Completed, Recommender System Study, Modu Research Institute

Dec 2020 - Jun 2021 Graduated, Alice AI Track (1st Cohort)

Sep 2020 - Nov 2020 Awarded First Place, University Capstone Design Competition Sep 2020 - Nov 2020

May 2017 Awarded First Place, SK Future Industry Technology Startup Competition

LANGUAGES

Korean (native)

English (conversational proficiency)
TOEIC 860, OPIC IM2

PROGRAMMING SKILLS

Python, C, R, HTML, CSS, Pytorch, SQL, Scikit-learn, Git, Redash

Publications

“K-ToPiC: Korean Dialect Toxicity Polishing with Curriculum Learning-based Span Detection”

Jungin Kim, Soohan Lim, Hyeseon Ahn, Sang-Min Choi, Yo-Sub Han

Proceedings of the Korea Computer Congress 2025 (KCC 2025).

“Marking Code Without Breaking It: Code Watermarking for Detecting LLM-Generated Code”

Jungin Kim, Shinwoo Park, Yo-Sub Han

Master’s Thesis, arXiv preprint arXiv:2502.18851, 2025 (Under Review). [[paper](#)]

“TCProF: Time-Complexity Prediction SSL Framework”

Joonghyuk Hahn, Hyeseon Ahn, Jungin Kim, Soohan Lim, Yo-Sub Han

In the Association for Computational Linguistics: NAACL 2025. [[paper](#)]

Publications

“CodeComplex: A Time-Complexity Dataset for Bilingual Source Codes”

Seung-Yeop Baik, Joonghyuk Hahn, Jungin Kim, Mingi Jeon, Yo-Sub Han, Sang-Ki Ko

arXiv preprint arXiv:2401.08719v2, 2024 (Under Review). [[paper](#)]

“SharedCon: Implicit hate speech detection using shared semantics”

Hyeseon Ahn, Youngwook Kim, Jungin Kim, Yo-Sub Han

In Findings of the Association for Computational Linguistics ACL 2024. [[paper](#)]

Patents

Code Watermarking Method and Apparatus that Preserves Syntax Application

No. 10-2025-0021267 | Filed: Feb 19, 2025

Code Complexity Prediction Method and Apparatus Using Semi-supervised Learning Application

No. 10-2024-0150149 | Filed: Oct 29, 2024

Projects

PROJECT INFORMATION

Project name	Period	Affiliation
Syntax-preserving Code Watermarking	Sep 2024 ~ Feb 2025	Yonsei University
Personalized News Recommendation Algorithm	Jan 2023 ~ Mar 2023	Alookso, Inc.
My Map: Personalized User Map	Mar 2022 ~ Jun 2022	Alookso, Inc
Officetel Price Prediction	Aug 2020 ~ Nov 2020	KAIST
COVID19: Twitter Sentiment Analysis	Mar 2020 ~ May 2020	Sangmyung University
Explainable Hate Speech Chatbot	Jun 2019 ~ Jul 2020	Sangmyung University
Analysis of the Causes of User Decline in North America	Aug 2017 ~ Sep 2017	Jocity. Inc.

Projects

Syntax-preserving Code Watermarking

#Code generation #Watermarking #LLM #Pytorch

[[Github](#), [Paper](#)]

- Addressed the issue where existing watermarking techniques caused execution or logic errors by modifying core grammatical elements of code (e.g., operators, keywords).
- Left grammatically important tokens untouched, and inserts watermarks only into non-syntax tokens

(a) Code Generation Prompt Write a function that sorts a list of numbers. If a number is even, it should appear before any odd numbers. Within each group (even/odd), the numbers should be sorted in ascending order. Use the following data as input: data = [5, 2, 9, 1, 4, 8, 7].	(b) Solution Code Without Watermarking <pre>def sort_even_before_odd(lst): evens = [num for num in lst if num % 2 == 0] odds = [num for num in lst if num % 2 != 0] result = sorted(evens) + sorted(odds) print(f"Sorted result: {result}") return result</pre> > Sorted result: [2, 4, 8, 1, 5, 7, 9]
(c) Code with SWEET Watermarking <pre>def sort_numbers(lst): even_n = [num for num in lst if num // 2 == 0] odd_n = [num for num in lst if num // 2 != 0] result = sorted(even_n) + sorted(odd_n) print(f"Sorted result: {result}") return result</pre> > Sorted result: [1, 2, 4, 5, 7, 8, 9]	(d) Code with STONE Watermarking <pre>def sort_numbers_even_first(arr): evens = [n for n in arr if n % 2 == 0] odds = [n for n in arr if n % 2 != 0] sorted_arr = sorted(evens) + sorted(odds) print(f"Sorted result: {sorted_arr}") return sorted_arr</pre> > Sorted result: [2, 4, 8, 1, 5, 7, 9]

Illustration of syntax-aware vs. syntax-agnostic watermarking techniques

Projects

Personalized News Recommendation Algorithm

#Python #SQL #Recommender System

- Constructed a user-item rating matrix based on user behavior logs (e.g., read time)
- Applied matrix factorization to learn latent user/item vectors.
- Predicted unseen item ratings and ranked them for personalized recommendation.
- Conducted iterative A/B testing and log analysis, resulting in an 8% increase in average user retention time.

Generate user-item rate matrix



Matrix factorization



Predict ratings for unseen items

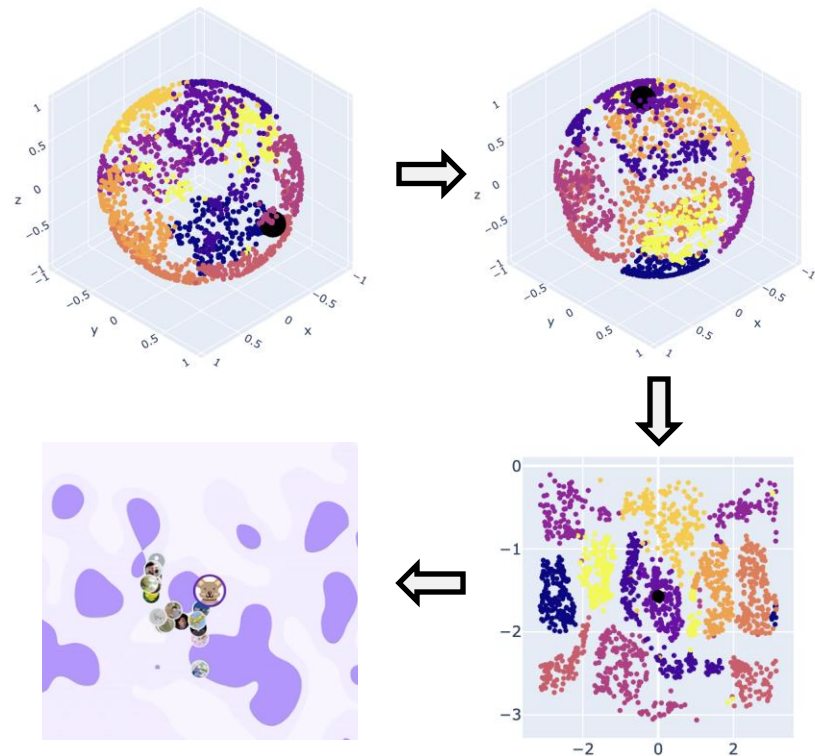
Workflow of the personalized recommendation algorithm using matrix factorization

Projects

My Map: Personalized User Map

#Python #SQL #UMAP

- User voting and interaction data are vectorized and projected onto a 3D sphere using UMAP.
- For each user, the sphere is rotated so that each user is positioned at the center and converted to 2D latitude-longitude coordinates for visualization.
- This process generates a personalized “My Map” for each user.



Personalized 2D map generated from 3D UMAP projection

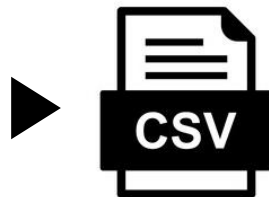
Projects

Officetel Price Prediction

#Python #Pandas #Numpy #Scikit-learn

- Developed an officetel price prediction model aimed at automating the traditional manual real estate appraisal process.
- Extracted key features from property appraisal reports that serve as the basis for price estimation.

구분	거래사례1	거래사례2	거래사례3
빌딩명	현대캐피탈빌딩	상촌까뮤빌딩	유진그림빌딩
전 경			
소재지	여의도동 15-21	여의도동 17-3	여의도동 24-3
규모	지상11층/지하5층	지상11층/지하3층	지상15층/지하3층
대지면적/용적율	4,248㎡/461.9%	5,044㎡/336.9%	1,974㎡/632.8%
건물면적 (용적율기준면적)	37,206.61㎡ (지상19,621.51㎡)	27,267.74㎡ (지상16,992.68㎡)	16,523.61㎡ (지상12,490.76㎡)
주구조	철근콘크리트구조	철근콘크리트구조	철근콘크리트구조
사용승인일	1992.08.08	1984.07.16	1981.08.05
거래시점	2013.03.05	2014.11.14	2015.02.16
매매가격(천원)	131,050,000	92,000,000	64,500,000
연면적기준 (원/㎡) (지상면적기준(원/㎡))	3,522,000 (6,679,000원/㎡)	3,374,000 (5,414,000원/㎡)	3,904,000 (5,164,000원/㎡)
연면적기준 (원/평) (지상면적기준(원/평))	11,644,000 (22,079,000)	11,154,000 (17,898,000)	12,906,000 (17,071,000)



Feature extraction process from real estate appraisal reports for price prediction

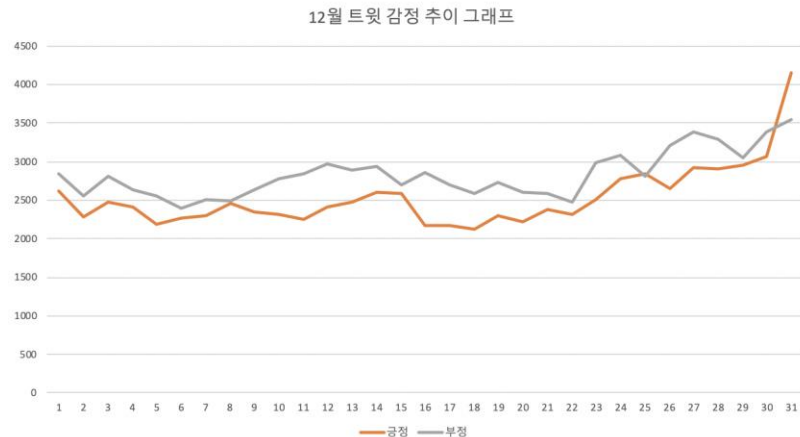
Projects

COVID19: Twitter Sentiment Analysis

#Python #Tensorflow #TwitterAPI #KoBERT

[Blog]

- Collected tweets from December to March for sentiment analysis.
- Used KoBERT to classify public sentiment on Twitter during the COVID-19 period.
- Conducted error analysis on 400 samples to identify model limitations.



Monthly sentiment trends on Twitter during
COVID-19 (example: december)

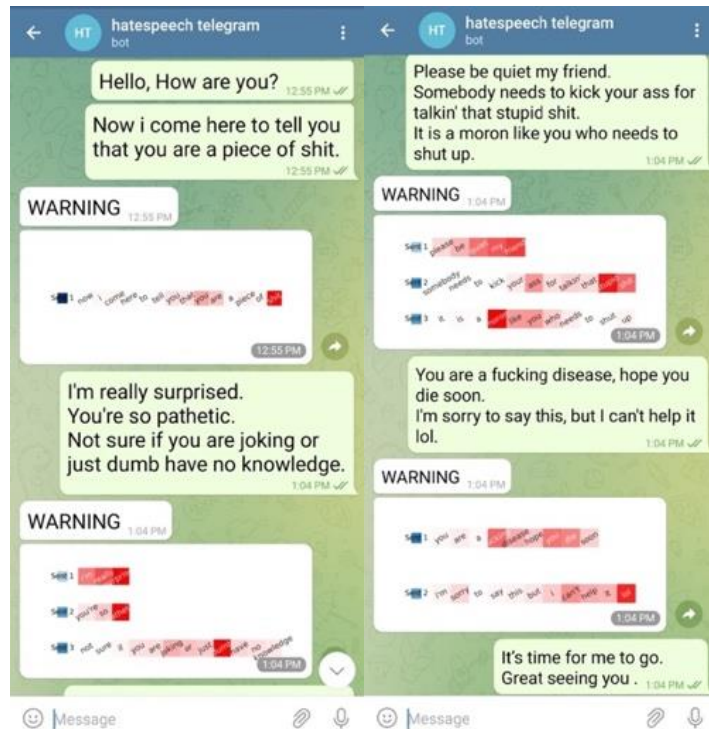
Projects

Explainable Hate Speech Chatbot

#NLP #Telegram bot API #Keras #Attention Mechanism

[[Github](#)]

- Explainable Hate Speech Detection using HAN (Hierarchical Attention Network)
- Implemented a chatbot that issues warnings and highlights offensive words or sentences based on attention weights during hate speech detection.



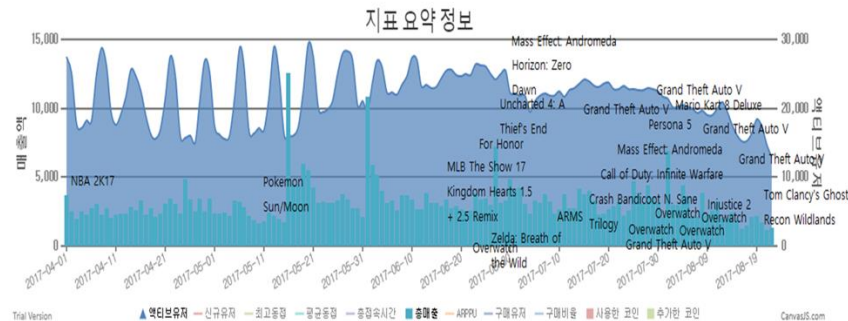
Telegram chatbot for hate speech detection with highlighted toxic spans via attention weights

Projects

Analysis of the Causes of User Decline in North America

#R #Data Analysis

- Analysis of the Decline in North American Users of *3on3 FreeStyle*
- Analyzed user trend and sales data
- Reviewed community feedback and competitor games



Correlation between user trends and competitor game release timeline

Thank You

Jungin Kim

+82(10) 9324-3486

Email: jungin3486@gmail.com

Seoul, Korea

[Google Scholar](#) ♦ [LinkedIn](#) ♦ [Github](#) ♦ [Blog](#)